19.11.18 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations
Nov – Dec 2018

Max. Marks: 100
Class: Third Year B.Tech.
Name of the Course: OPERATING SYSTEMS
Course Code: **UCEC501**

Duration: 3 Hours
Semester: V
Branch: COMP

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	What is a Monolithic structure? What is a Microkernel? Draw neat diagrams and give example for each.	10
Q 1 (b)	Write a short note on – Shell. OR Write a short note on – Android Operating System.	10
Q2 (a)	What is a thread? What are types of threads? Draw and explain multithreaded process model.	10
Q2 (b)	What is a process? Draw and explain five-state process model. OR What is a PCB? Draw and explain components of PCB.	10
Q3 (a)	Describe short term, medium term and long term scheduling with neat diagram.	10
Q3 (b)	What is preemptive and non-preemptive scheduling? Compare different process scheduling algorithms. OR There are 5 processes A to E which are waiting to be scheduled. Their arrival times are 0,1,3,9 & 12 Second respectively and their processing times are 3,5,2,5 and 5 Seconds respectively. What is the average turn-around time using FCFS, SJF and Round-Robin (with a quantum of 1 Second) scheduling?	10
Q4 (a)	Explain Producer Consumer problem. and Solve it using any one method.	10

Q4 (b)	Explain deadlock avoidance strategy for system having single instance of all resources and for system having multiple instances of all resources. OR What is inter-process communication? With the help of neat diagrams explain shared memory and message passing techniques.	10
Q5 (a)	What are the categories of I/O devices? Explain operating system I/O design issues.	10
Q5 (b)	What criteria should be adopted for choosing type of file organization? Describe the implementation of file allocation techniques. OR Suppose that a disk drive has 200 tracks, numbered 0 to 199. The head of moving disk is currently serving a request at track 143, The queue of pending requests in FIFO order is 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total number of head movement needed to satisfy all the pending requests for following disk scheduling algorithm viz. FCFS, SSTF and SCAN (direction → increasing order) ?	10

22.11.2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
Nov – Dec 2018

Max. Marks: 100 | **Duration: 3hours**
Class: T.E. | Semester: V
Name of the Course: Data Networks
Branch: Computer | Course Code: UCEC502

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Explain implication of wireless medium and mobility on protocol stack design .	10
Q 1 (b)	Identify the limitation of MAC layer protocols of wired network when applied to mobile networks.	10
Q2 (a)	Explain the IP Subnetting with example	10
Q2 (b)	Why IPV6 addressing is better than IPV4 justify and explain in detail.	10
Q3 (a)	What is MANET? Explain DSDV Protocol. OR What is multicast routing? Explain MOSPF.	10
Q3 (b)	OR What are the services provided by the network layer.Explain link state routing algorithm	10
Q4 (a)	Is it possible for an application to enjoy reliable data transfer even when the application runs over UDP? If so, how? OR Discuss the four way handshake for TCP connection termination	10
Q4 (b)	Explain mobile TCP. Give the advantage & disadvantage of Mobile TCP.	10
Q5 (a)	Write short notes on any two. 1) HTTP 2) DNS 3) WAP 4) WWW	10
Q5 (b)	Write short notes on any two. 1) FTP 2) SMTP 3) Telnet 4) NAT	10

Q.3(b) Routing table of R1 is shown as below:

Mask	Network Address	Next Hop	Interface
/27	123.10.10.0	---	M0
/24	199.17.25.0	132.15.15.17	M1
/16	146.19.0.0	132.15.15.17	M1
/16	210.36.0.0	---	M2
/16	132.15.0.0	---	M1
Default	Default	210.36.18.10	M2

Show the forwarding process, if packet is arrived at router R1 with destination address 146.19.15.10

26.11.2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
Nov – Dec 2018

Max. Marks: 100
Class: T.Y.B.Tech
Name of the Course: Theory of Computer Science
Course Code: UCEC503

Duration: 3 Hrs
Semester: V
Branch: Comp

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Construct NFA with ε moves for regular expression (a+b)*aba(a+b)* and obtain DFA	10
Q 1 (b)	State and prove pumping lemma for regular languages and prove that L={0^{2n}|n>=1} is not regular language. OR	10
Q1 (b)	Design DFA to accept following languages over the input alphabet{0,1} 1) {w|w starts with 0 and has odd length or w starts with 1 and has even length} 2) {w|every odd position of w is 1}	10
Q2 (a)	Convert the following grammar to CNF S→abAB ,A→bAB|ε ,B→Baa|A| ε OR	10
Q 2 (a)	Find Greibach normal form equivalent to the following CFG A → Bb|BB ,B →AB|a ,S →BA |ab	10
Q 2 (b)	Obtain leftmost, rightmost derivation and derivation tree for the string "cccbaccba" .The grammar is S →SSa|SSb|c	10
Q3 (a)	Construct PDA from the grammar over∑(a,b) E→E+E|E-E|(E) |id id→a|b	10
Q3 (b)	Design NPDA for CFG S→aAA|a A→bS|aS OR	10
Q3 (b)	Design PDA for the language L={wcw ^{r} |w ε(a,b)* and w ^{r} is reverse of given string w}	10

Q4 (a)	Design TM for the language $L=\{a^n b^n c^n \mid n>=1\}$ OR	10
Q4 (a)	Design TM for recognizing the language $L=\{x \mid x \in (a,b)^* \& n_{a(x)}=n_{b(x)}$	10
Q4 (b)	Write short note on (each with 5 marks) 1) Halting Problem of TM Variants of Turing machine 2) Universal Turing Machine	10
Q5 (a)	Explain the undecidability of PCP?Does PCP with following two list _ A=(10,011,101) and B=(101,11,011) have a solution? Justify your answer.	10
Q5 (b)	Write short note on (each with 5 marks) 1)Halting Problem of TM 2) Properties of Recursive and Recursively Enumerable Languages OR	10
Q5 (b)	1) Design TM for the language $L=\{a^n b^n \mid n>=1\}$ 2) Explain Closure properties of CFG	10

28-11-18 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov-Dec 2018

Max. Marks: 100 | **Duration: 3 Hrs**
Class: T.Y. | **Semester: V**
Name of the Course: Advanced Database Management System | **Branch: Computer**
Course Code: UCEC504

Instructions:
(1) **All Questions are Compulsory**
(2) **Draw neat diagrams**
(3) **Assume suitable data if necessary**

Question No.		Max. Marks
Q.1 (a)	What are the strategies used for View integration, explain with example? OR Explain the role of information system in organization.	10
Q.1 (b)	Explain Index tuning and Query tuning with example.	10
Q.2 (a)	Consider a disk with block size B=512 bytes. A block pointer is P=6 bytes long and a record pointer is P R =7 bytes long. A file has r=30,000 EMPLOYEE records of fixed-length. Each record has the following fields: NAME (30 bytes), SSN (9bytes), DEPARTMENTCODE (9 bytes), ADDRESS (40 bytes), PHONE (9 bytes),BIRTHDATE (8 bytes), SEX (1 byte), JOBCODE (4 bytes), SALARY (4 bytes, real number). An additional byte is used as a deletion marker. 1.Calculate the blocking factor bfr and the number of file blocks b assuming an unspanned organization. (4) 2.Suppose the file is ordered by the key field SSN and we want to construct a primary index on SSN. Calculate the following : (i) the index blocking factor bfr i (which is also the index fan-out fo). (2) (ii) the number of first-level index entries and the number of first-level index blocks. (4) OR Consider an example and explain implementation of Dynamic hashing for Indexing with example and give overflow handling technique for dynamic hashing.	10
Q.2 (b)	What is B-tree indexing? How it is different than the ordered indexing methods? Implement B-tree indexing on the given database on	10

"EMPNO". order of the tree node, n = 3

EMPNO	FIRSTNME	MiD INIT	LASTNAME	WORK DEPT	PHONENO	HIREDATE
000010	CHRISTINE	I	HAAS	A00	3978	1965-01-01
000020	MICHAEL	L	THOMPSON	B01	3476	1973-10-10
000030	SALLY	A	KWAN	C01	4738	1975-04-05
000050	JOHN	B	GEYER	E01	6789	1949-08-17
000060	IRVING	F	STERN	D11	6423	1973-09-14
000070	EVA	D	PULASKI	D21	7831	1980-09-30
000090	EILEEN	W	HENDERSON	E11	5498	1970-08-15
000100	THEODORE	Q	SPENSER	E21	0972	1980-06-19
000110	VINCENZO	G	LUCCHESSI	A00	3490	1958-05-16

Q.3 (a) (10)

Consider these relations with the following properties:

r(A, B, C)	s(C, D, E)
30,000 tuples	60,000 tuples
25 tuples fit on 1 block	30 tuples fit on 1 block

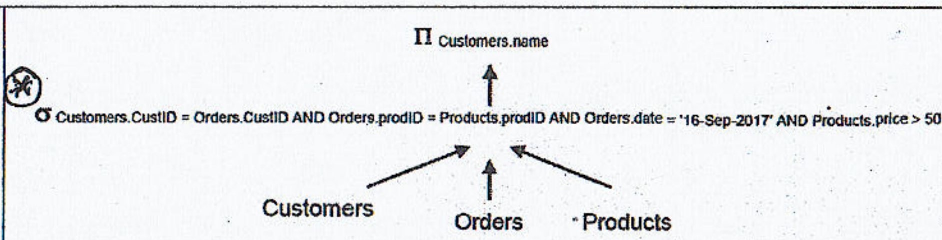
a) Estimate the number of disk block accesses required for a natural join of r and s using a nested-loop join if r is used as the outer relation.

b) Estimate the number of disk block accesses required for a natural join of r and s using a block nested-loop join if s is used as the outer relation. Assume that there are more than 2000 memory buffers available to facilitate this operation, where each memory buffer can buffer one disk block.

Q.3 (b) (10)

Given the following three linked tables:
Customers (custID, name, country)
Products (prodID, price)
Orders (orderID, custID*, prodID*, date)
Suppose we have the following query and its corresponding initial parse tree:

SELECT Customers.name FROM Customers, Orders, Products WHERE Customers.custID = Orders.custID AND Orders.prodID = Products.prodID AND Orders.date = '16-Sep-2017' AND Products.price > 50;

	 (i) What problems arise if the query is executed based on the above parse tree? (ii) Transform the above parse tree into one that corresponds to the most efficient way of processing the query.	
Q 4 (a)	With the help of example explain how object oriented data model is different than Relational model OR What are the various concurrency control techniques? Compare Lock based Concurrency Control strategies in detail.	10
Q 4 (b)	Explain interquery and intraquery Parallelism in parallel Database with example. OR What is horizontal and vertical fragmentation? What are the types of horizontal fragmentation. Perform horizontal Fragmentation for student relation as given below. Students (studentrollno., Student Name, Course Name, Course Name, Course fees, year)	10
Q.5 (a)	Explain the distinction between 'Semi-structured data' as exhibited by data description languages such as XML and 'Structured data' as exhibited by the relational model.	10
Q.5 (b)	What do you understand by XML Schema? What is the difference between attributes and elements in XML? List some of the important attributes used to specify elements in XML schema. OR Write a short note on Xpath and Xquery with the help of suitable example.	10

(*) σ customer.custID = orders.CustID AND orders.ProdID = Products.ProdID AND orders.Date = '16-Sept-2017'

30.11.2018 (E)

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End Semester Exam
November – December 2018

Max. Marks: 100 | **Duration: 3hrs**
Class: TY | Semester: V
Name of the Course: Software Engineering | Branch: Computer Engineering
Course Code: **UCEC505**

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Write a note on Open Source Model.	5M
Q 1 (b)	Write a note on Feature Driven development.	5M
Q1 (c)	State advantages and disadvantages of Spiral model	5M
	OR	
Q1 (c)	Differentiate between Waterfall and RAD model. (5 points)	5M
Q1 (d)	Write a note Earned Value Analysis.	5M
Q2 (a)	Draw class diagram for Hotel Management System.	10M
	OR	
Q2(a)	Draw Use Case Diagram for any two'scenarios of Banking System.	10M
Q.2 (b)	Draw Sequence Diagram for Library Management System.	10M
3 (a)	Write a note on Pattern Based Software Engineering.	10M
	OR	
Q3 (a)	Write a note on design principles.	10M

Q3 (b)	Explain User Interface Design rules in detail.	10M
Q4 (a)	Describe Change Control for software with suitable example.	10M
Q4 (b)	Explain three refactoring types in mapping model to code with suitable code for required part of the example.	10M
	OR	
Q4 (b)	Explain mapping associations to collection in mapping model to code with suitable code for required part of the example.	10M
Q5 (a)	Consider a banking system that computes rate of interest based on balance amount in Savings Account according to the following rules: • If Account Balance is between Rs.10000 and Rs.39500, valid for rate of interest of 2%. • If Account Balance is between Rs.39501 and Rs.60500, valid for rate of interest of 4%. • If Account Balance is between Rs.60501 and Rs.1 lakh and above, valid for rate of interest of 6%. **Design 5 test cases covering all valid and invalid conditions for the above example using ECP analysis.**	10M
	OR	
Q5 (a)	Explain BVA with the help of suitable test case example.	10M
Q5 (b)	Explain OO testing methodology in detail.	10M